

Practical 1 Tasks

1. PING Command

Objective

To test connectivity between two hosts and measure network response time.

Exercise Questions

1. Write the syntax of the ping command in Windows and Linux.
2. Ping the localhost and record the observations.
3. Ping your default gateway and note the packet loss and RTT.
4. Ping the domain google.com. What does the command verify?
5. Explain the significance of TTL in ping results.
6. Execute continuous ping and stop it manually. Write the command used.
7. Compare ping results for a local IP address and a remote website.
8. What happens when you ping an unreachable host?
9. Use ping with packet size modification and record the output.
10. Differentiate between ICMP Echo Request and Echo Reply.

2. IPCONFIG / IFCONFIG Command

Objective

To view and analyze network interface configuration.

Exercise Questions

1. Display the IP configuration of your system using ipconfig or ifconfig.
2. Identify:
 - IPv4 Address
 - Subnet Mask
 - Default Gateway
 - MAC Address
3. Explain the purpose of DHCP.
4. Use ipconfig /all and list additional information obtained.
5. Release and renew the IP address using appropriate commands.
6. Differentiate between static and dynamic IP addressing.
7. What is the purpose of loopback address?
8. Compare outputs of ifconfig and ip addr.
9. Identify active and inactive interfaces.
10. Explain the significance of subnet mask in communication.

3. IWCONFIG Command

Objective

To examine wireless network interface parameters.

Exercise Questions

1. Display wireless interface details using `iwconfig`.
2. Identify the ESSID and operating frequency.
3. What is the significance of signal level and link quality?
4. Explain Managed and Monitor modes.
5. Change the wireless mode using `iwconfig`.
6. Observe bitrate and transmission power values.
7. Compare wired and wireless interface configurations.
8. Explain the purpose of Access Point MAC address.
9. What happens when ESSID is off/any?
10. List limitations of wireless communication observed from the output.

4. NETSTAT Command

Objective

To inspect active network connections and ports.

Exercise Questions

1. Display all active connections using `netstat`.
2. Identify listening ports on the system.
3. Explain the meaning of TCP states:
 - LISTENING
 - ESTABLISHED
 - TIME_WAIT
4. Find the process ID associated with a port.
5. Display routing table using `netstat` options.
6. Compare TCP and UDP entries.
7. Identify foreign and local addresses.
8. Explain why some ports remain in TIME_WAIT state.
9. Display statistics of network protocols.
10. What security risks can be identified using `netstat`?

5. TRACEROUTE / TRACERT Command

Objective

To determine the path taken by packets to a destination.

Exercise Questions

1. Trace the route to `google.com`.

2. Record the number of hops observed.
3. Explain the significance of each hop.
4. Why do some hops display * * *?
5. Compare ping and traceroute commands.
6. Explain the role of TTL in traceroute.
7. Identify the ISP router from the output.
8. Trace a route to an unreachable destination.
9. Measure delay variation between hops.
10. Explain practical applications of traceroute.

6. ROUTE Command

Objective

To view and manipulate routing tables.

Exercise Questions

1. Display the routing table using route print or route.
2. Identify the default route in the table.
3. Explain:
 - Destination
 - Gateway
 - Genmask
 - Metric
4. Add a temporary route using the route command.
5. Delete a route from the routing table.
6. Differentiate between static and dynamic routing.
7. Explain the significance of routing metrics.
8. Observe changes after adding a route.
9. What is the role of gateway in routing?
10. Explain how routing tables help packet forwarding.

7. NSLOOKUP Command

Objective

To perform DNS queries and domain name resolution.

Exercise Questions

1. Find the IP address of google.com.
2. Query the mail server records of a domain.
3. Explain forward and reverse DNS lookup.
4. Find the authoritative DNS server of a domain.
5. Compare nslookup and ping outputs.

6. Query using a specific DNS server.
7. What happens when DNS resolution fails?
8. Explain the purpose of DNS caching.
9. Perform reverse lookup for a given IP address.
10. Identify different DNS record types.

8. HOSTNAME Command

Objective

To identify and manage system hostnames.

Exercise Questions

1. Display the hostname of your machine.
2. Explain the role of hostname in networking.
3. Differentiate hostname and domain name.
4. Change the hostname temporarily.
5. Verify hostname resolution using ping.
6. Explain FQDN (Fully Qualified Domain Name).
7. How does hostname assist in network administration?
8. Observe hostname entry in configuration files.
9. What issues arise from duplicate hostnames?
10. Compare hostname usage in Windows and Linux.

9. DIG Command

Objective

To perform detailed DNS queries.

Exercise Questions

1. Query A records of a domain using dig.
2. Query MX records and explain their purpose.
3. Compare outputs of dig and nslookup.
4. Perform reverse DNS lookup using dig.
5. Identify TTL values in DNS responses.
6. Explain DNS header sections in dig output.
7. Query NS records of a domain.
8. Trace DNS resolution using dig options.
9. What is recursion in DNS queries?
10. Explain practical applications of dig.

10. TELNET Command

Objective

To test remote connectivity and service availability.

Exercise Questions

1. Connect to a remote host using telnet.
2. Test connectivity to port 80 of a web server.
3. Explain why telnet is considered insecure.
4. Compare telnet and SSH.
5. Observe server response after connection.
6. What happens when a port is closed?
7. Use telnet to manually send HTTP requests.
8. Explain plaintext transmission in telnet.
9. Identify scenarios where telnet is still useful.
10. Record observations for successful and failed connections.

11. SSH Command

Objective

To securely access remote systems.

Exercise Questions

1. Establish SSH connection to a remote host.
2. Explain the role of encryption in SSH.
3. Generate SSH keys and explain their purpose.
4. Compare password authentication and key authentication.
5. Copy files securely using SCP.
6. Explain the role of port 22.
7. What happens when host key verification fails?
8. Compare SSH and Telnet security.
9. Identify common SSH applications in networking.
10. Observe SSH session establishment steps.

12. ARP Command

Objective

To examine Address Resolution Protocol tables.

Exercise Questions

1. Display the ARP cache using arp command.
2. Identify IP and MAC address mappings.
3. Explain the role of ARP in LAN communication.
4. Clear the ARP cache and observe changes.
5. Add a static ARP entry.
6. Differentiate static and dynamic ARP entries.
7. Explain ARP request and ARP reply.
8. What is Gratuitous ARP?
9. Explain ARP spoofing attacks.
10. Observe changes after pinging a new host.

13. NMAP Command

Objective

To scan hosts and discover open ports/services.

Exercise Questions

1. Scan a host for open ports.
2. Identify running services using nmap.
3. Perform TCP SYN scan.
4. Explain stealth scanning.
5. Compare TCP and UDP scanning.
6. Detect operating system using nmap.
7. Scan an entire subnet.
8. Explain the significance of filtered ports.
9. Perform version detection scan.
10. Discuss ethical considerations of port scanning.

14. GETMAC Command

Objective

To identify MAC addresses of network adapters.

Exercise Questions

1. Display MAC addresses using getmac.
2. Compare physical and logical addressing.
3. Identify transport name associated with adapters.
4. Explain uniqueness of MAC addresses.
5. Compare MAC address and IP address.
6. Record MAC addresses of all interfaces.
7. Explain vendor identification from MAC addresses.
8. Why are MAC addresses important in LANs?

9. Compare outputs of getmac and ipconfig.
10. Explain MAC spoofing.

15. PATHPING Command

Objective

To analyze network path and packet loss.

Exercise Questions

1. Execute pathping to a remote server.
2. Compare pathping and tracert outputs.
3. Identify packet loss percentage at each hop.
4. Explain latency measurements.
5. Determine the problematic hop in a route.
6. Why does pathping take longer to execute?
7. Compare ping, tracert, and pathping.
8. Explain route stability analysis.
9. Identify congestion using pathping results.
10. Record observations for a long-distance destination.

16. SS Command

Objective

To investigate socket statistics in Linux.

Exercise Questions

1. Display active TCP connections using ss.
2. Compare ss and netstat commands.
3. Display listening ports using ss.
4. Filter connections by protocol.
5. Identify process associated with a socket.
6. Explain socket states observed.
7. Display summary statistics using ss.
8. Explain advantages of ss over netstat.
9. Observe IPv4 and IPv6 socket entries.
10. Identify high network usage processes.

17. ROUTE ADD Command

Objective

To add static routes manually.

Exercise Questions

1. Add a static route to a network.
2. Verify the route addition using routing table.
3. Delete the added route.
4. Explain gateway selection in route add.
5. Compare persistent and temporary routes.
6. Add a route with metric value.
7. Observe packet forwarding after route addition.
8. What errors occur with invalid gateways?
9. Explain practical applications of static routes.
10. Differentiate route add and dynamic routing protocols.

Viva Questions

1. What is the difference between logical and physical addressing?
2. Explain the role of DNS in networking.
3. Why is ICMP important?
4. What is the purpose of routing tables?
5. Explain packet loss and latency.
6. Compare TCP and UDP.
7. What are open and closed ports?
8. Explain SSH tunneling.
9. What is ARP poisoning?
10. Why are network diagnostic tools important?

Advanced Tasks

1. Analyse packet delivery, latency, routing path, and packet loss between hosts.
2. Perform end-to-end troubleshooting of a simulated network failure for the scenario where a user is not able to access a webpage.